EMMANUEL JOSEPH P. SIBAYAN

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Education

Mapua Malayan Colleges Laguna Cabuyao, Laguna
Bachelor of Science in Information Technology, Specialized in Cybersecurity August 2026
President's List (2022-2023), Dean's List (2022-2026)

Professional Experience

Freelance Data Specialist (Tax Compliance) January 2022 - Present
Remote (Quarterly)

- Managed quarterly data encoding and BIR compliance filing for multiple corporate clients, ensuring 100% accuracy in financial reporting.
- Identified significant bottlenecks in manual VAT data entry, serving as the primary subject matter expert for the development of the Automated Tax Compliance Utility.
- Ensured strict adherence to the Data Privacy Act (RA 10173) while handling sensitive corporate financial records.

Technical Skills and Projects

Automated Tax Compliance Utility January 2026 - Present
Full Stack Developer

- Architecting a local-first automation tool using Python to streamline the encoding of VAT-related data into legacy government systems, reducing manual entry time by an estimated 80%.
- Developing a robust data-cleansing pipeline using pandas and Regular Expressions (Regex) to sanitize "dirty" client data, ensuring 100% compliance with alphanumeric and TIN format requirements.
- Enforcing Data Privacy (RA 10173) by designing a strictly offline architecture, ensuring sensitive financial records are processed entirely in-memory without cloud dependencies.

Streamline: A Water Quality Monitoring System August 2025 – July 2026
Full Stack Developer

- Architected and deployed a production-grade monitoring system for the Laguna Lake Development Authority (LLDA) using Django 6 and React 19, handling real-time IoT data from river networks.
- Developed an IoT ingestion pipeline with HMAC payload signing and MAC address binding to ensure secure data transmission from Heltec LoRa P2P devices.
- Implemented advanced security protocols, including 2FA, JWTs stored in HttpOnly cookies, and Redis-backed rate limiting to protect sensitive environmental data.
- Automated compliance reporting by integrating a headless browser subprocess (Playwright) to capture dynamic map views and generate PDF reports stored in S3-compatible MinIO storage.

An AES- Based Cryptographic Model with DNA Encoding and Chaos Mapping November 2025
Encryption Model Developer

- Developed a modular encryption pipeline using Python and the PyCryptodome library, implementing AES-256 in Cipher Block Chaining (CBC) mode.
- Engineered a custom transposition layer using chaotic mathematical functions to generate key-dependent pseudo-random permutation sequences.
- Optimized data structures for DNA nucleotide mapping, ensuring full algorithmic reversibility for seamless decryption.

E-Rehistro: A Voter Registration System April 2024
Backend Developer

- Developed a high-performance web application using ASP.NET and C# to digitize and automate voter registration processes.
- Designed and integrated a SQL Server database using Entity Framework (EF Core), utilizing LINQ queries to perform efficient CRUD operations and data filtering.
- Built server-side validation logic to enforce eligibility requirements, ensuring data accuracy and preventing invalid registration entries.

Achievements & Certifications

CompTIA Tech+ Certificate December 2025
Google Cloud Computing Foundations Certificate February 2025

Skills & Interests

Programming & Frameworks: Python, C#, JavaScript/TypeScript, PHP, C++, Java
Database & Geospatial: PostgreSQL, PostGIS, SQL Server, Redis, GeoDjango, MapLibre GL
Tools & Frameworks: Django, React, pandas, PyAutoGUI, Playwright, MinIO, Git
Language: English (C1), Filipino (Native)
Interests: Project Management, System Development, Cybersecurity, Cloud Computing, PC Hardware Building, Music